CHAPTER 2

RELATIONS AND FUNCTIONS

CHAPTER ESSENCE AT A GLANCE:
Relations pair inputs to outputs; a Function is a disciplined relation where EVERY input has exactly ONE unique output (no single x points to two different y values!). Graphing basic functions like |x| and [x] reveals their domain and range instantly!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Cartesian Product A × B:** Set of all ordered pairs (a, b) with a A and b B. If n(A) = p and n(B) = q, then n(A × B) = pq, and total relations from A to B = 2^(pq).

• **Relation vs Function:** A relation is a function if and only if EVERY element of domain A has exactly ONE unique output in B (no element maps to multiple images; no domain element is left unmapped).

• **Modulus Function f(x) = |x|:** Defined as x for x ≥ 0 and -x for x < 0. Domain = R (all reals), Range = [0, ∞) (non-negative reals). Graph is symmetric V-shape at origin.

• **Signum Function sgn(x):** Defined as 1 for x > 0, 0 for x = 0, and -1 for x < 0. Domain = R, Range = {-1, 0, 1}.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$n(A \times B) = p \cdot q \rightarrow \text{Total Relations} = 2^{p \cdot q}$$

Cartesian Product and Relation Multiplicity

$$f(x) = |x| = +x \ (x \geq 0), \ -x \ (x < 0) \quad (\text{Domain} = \mathbf{R}, \ \text{Range} = [0, \infty))$$

Modulus Function Definition and Interval Range

$$\text{sgn}(x) = +1 \ (x > 0), \ 0 \ (x = 0), \ -1 \ (x < 0) \quad (\text{Range} = \{-1, 0, 1\})$$

Signum Function Piecewise Definition

$$f(x) = [x] = n \quad (n \leq x < n + 1, \ n \in \mathbf{Z})$$

Greatest Integer Step Function Formulation

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** A relation is NOT a function if an input x has two arrows leaving it, or if any element in domain A has no partner in B!

• **Alert:** Vertical Line Test: If any vertical line intersects a graph in MORE than one point, the graph is NOT a function!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define a relation R from set A to set B. What is the total number of relations from A to B if n(A) = p and n(B) = q? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Definition]** A relation R from a non-empty set A to a non-empty set B is any subset of the Cartesian product A × B (i.e. R subset of A × B). [1 Mark]

• **[Step 2: Total Number of Relations]** Since n(A × B) = p x q, the total number of subsets of A × B is 2^(p x q). Total relations = 2^(p q). [1 Mark]

• **[Boxed Result: Total number of relations = 2^(p q)]**

2. Let A = {1, 2, 3, 4, 5, 6}. Define a relation R on A by R = {(x, y) : y = x + 1}. Write R in roster form, and find its domain and range. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Roster Form]** R = {(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)}. [Notice (6, 7) not in R as 7 not in A]. [1 Mark]

• **[Step 2: Domain and Range]** Domain = Set of all first elements = {1, 2, 3, 4, 5}. Range = Set of all second elements = {2, 3, 4, 5, 6}. [1 Mark]

• **[Boxed Result: Domain = {1, 2, 3, 4, 5}, Range = {2, 3, 4, 5, 6}]**

3. Draw the graph of the Modulus Function f(x) = |x|. State its domain and range. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Definition & Splitting]** f(x) = |x| = { x if x ≥ 0; -x if x < 0 }. [1 Mark]

• **[Step 2: Graph Description]** The graph consists of two rays originating from the origin (0, 0): the line y = x in the first quadrant and the line y = -x in the second quadrant, forming a symmetric V-shape. [1 Mark]

• **[Step 3: Domain and Range]** Domain = Set of all real numbers R = (-∞, ∞). Range = Set of non-negative real numbers [0, ∞). [1 Mark]

• **[Boxed Result: V-shaped graph; Domain = R, Range = [0, ∞)]**

4. Draw the graph of the Signum Function. Write its definition, domain, and range. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Definition of Signum Function]** sgn(x) = { 1 if x > 0; 0 if x = 0; -1 if x < 0 } (or f(x) = |x| over x for x ≠ 0 and f(0) = 0). [1 Mark]

• **[Step 2: Graph Description]** Consists of a horizontal ray y = 1 for x > 0 (with an open circle at (0, 1)), an isolated point at (0, 0), and a horizontal ray y = -1 for x < 0 (with an open circle at (0, -1)). [1 Mark]

• **[Step 3: Domain and Range]** Domain = R (all real numbers). Range = {-1, 0, 1} (a discrete set of 3 numbers). [1 Mark]

• **[Boxed Result: Domain = R, Range = {-1, 0, 1}]**

5. Draw the graph of the Greatest Integer Function f(x) = [x]. State its domain and range. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Definition]** [x] denotes the greatest integer less than or equal to x (e.g. [2.3] = 2, [-1.5] = -2). [1 Mark]

• **[Step 2: Graph (Step Function)]** The graph is a series of horizontal steps: y = 0 for 0 ≤ x < 1; y = 1 for 1 ≤ x < 2; y = -1 for -1 ≤ x < 0. Each step has a closed left endpoint and an open right circle, resembling a staircase. [1 Mark]

• **[Step 3: Domain and Range]** Domain = R (all real numbers). Range = Z (set of all integers). [1 Mark]

• **[Boxed Result: Step graph; Domain = R, Range = Z (integers)]**

6. Let $f(x) = x^2$ and $g(x) = 2x + 1$ be two real functions. Find: (a) (f + g)(x) (b) (f - g)(x) (c) (f g)(x) (d) (f over g)(x) and state its domain. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Addition and Subtraction]** (a) $(f + g)(x) = f(x) + g(x) = x^2 + 2x + 1$. [1 Mark]

• (b) $(f - g)(x) = f(x) - g(x) = x^2 - (2x + 1) = x^2 - 2x - 1$. [1 Mark]

• **[Step 2: Multiplication and Division]** (c) $(f g)(x) = f(x) . g(x) = x^2 (2x + 1) = 2x^3 + x^2$. [1 Mark]

• (d) (f over g)(x) = f(x) over g(x) = x^2 over (2x + 1). Defined when 2x + 1 != 0 => x != -1 over 2. Domain = R - {-1 over 2}. [1 Mark]

• **[Boxed Result: (f over g)(x) = x^2 over (2x + 1), Domain = R - {-1 over 2}]**

7. (a) Find the domain and range of the function f(x) = |x - 1|. (2 Marks)n(b) Find the domain and range of the function f(x) = 1 over √(x - 2). (2 Marks) [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Function f(x) = |x - 1|]** Defined for all real numbers: Domain = R. Since absolute value is non-negative and reaches 0 at x = 1: Range = [0, ∞). [2 Marks]

• **[Step 2: Function f(x) = 1 over √(x - 2)]** For square root in denominator: x - 2 > 0 (strictly positive to avoid division by zero): x > 2. Domain = (2, ∞). As x → 2+, f(x) → ∞; as x → ∞, f(x) → 0+. Range = (0, ∞). [2 Marks]

• **[Boxed Result: |x-1|: Dom=R, Rng=[0, inf); 1 over √(x-2): Dom=(2, inf), Rng=(0, inf)]**